

Why Tools Matter

Todd Austin
University of Michigan

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■ A long time ago, in a place far away...

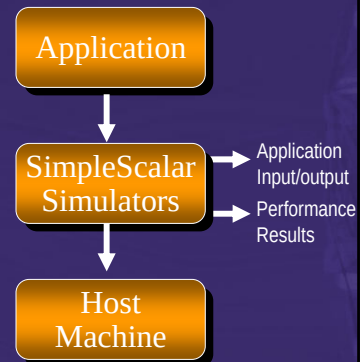


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My Experience: SimpleScalar Tool Set

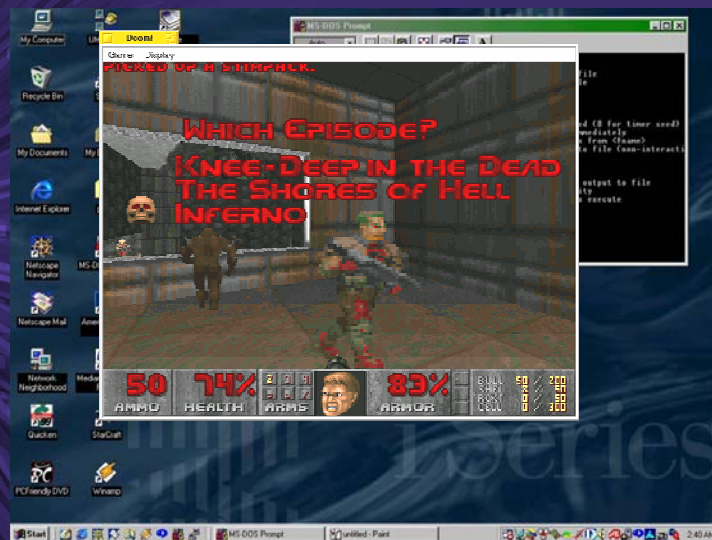
- Computer system design and analysis infrastructure
 - Uniprocessor models
 - User-mode input/output
 - Supports many ISAs
 - Portable to host platforms
 - Deployed widely in academia and industry for more than a decade
- Available with source and docs from www.SimpleScalar.com



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Obligatory Shot of Simulator Running DOOM



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Reasons Against Building & Releasing Tools

- Don't want to loose your "competitive edge"
- Often, tools development is not viewed as "research"
- User support and code maintenance is very demanding, couldn't your time be spent better?
- Broken tools could be harmful to the state-of-the-art
- Vast majority of user feedback is negative

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Goals of This Talk

Encourage aspiring tools hackers to build and release their tools.

Give you my best advice for undertaking such a challenge.

Suggest some promising (simulator) tool directions to pursue.

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■ Why Tools Matter -- to the Community

The altruistic viewpoint...

- Accelerate the pace of research
 - Good tools enable researchers
- Accelerate the pace of tool development
 - Through enabling infrastructure development
 - Through the growth of a community of developers

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■ Why Tools Matter -- to Me

The “narcissistic” viewpoint...

- Amplify your contribution to the research community
 - Take credit for other peoples research!
- Gain public recognition

- sim-safe: SimpleScalar/PISA Tool Set version 3.0 of August, 2003.
Copyright (c) 1994-2003 by **Todd M. Austin**, Ph.D. and SimpleScalar, LLC.
All Rights Reserved. This version of SimpleScalar is licensed for academic non-commercial use. No portion of this work may be used by any commercial entity, or for any commercial purpose, without the prior written permission of SimpleScalar, LLC (info@simplescalar.com).

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■ Advice for Building Successful Tools

Based on my experiences with SimpleScalar.

Plus, observations from other tool developers.

But don't forget... you'll need a lot of luck, too!

And now... six pieces of advice.

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■ Advice: Let Research Drive the Outcome

- Tools serve research demands
 - Let your research agenda define these demands
- Great infrastructure enables research
 - Build the infrastructure and you will get there first!
 - E.g., SimpleScalar was first to recognize the positive benefits of mis-speculation [PAID'95]
- It will sharpen the focus and usability of the infrastructure



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■ Advice: Be Innovative

- Don't try to only improve engineering, re-imagine the design
- You will stay interested in the project longer
- It will become easier to publish your work



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■ Innovation: Execution-Driven Simulation

- Trace-based simulation
 - Simulator reads instruction “trace”
 - Simple to implement, no functional component needed
- Execution-driven simulation
 - Simulator “runs” the program, generating complete execution
 - More difficult to implement, requires instruction function and I/O handling
 - Approach has many advantages, e.g., access to values, misspeculation

Inst Trace



H/W
model

Program



H/W
model

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■ Innovation: External I/O Tracing (EIO)

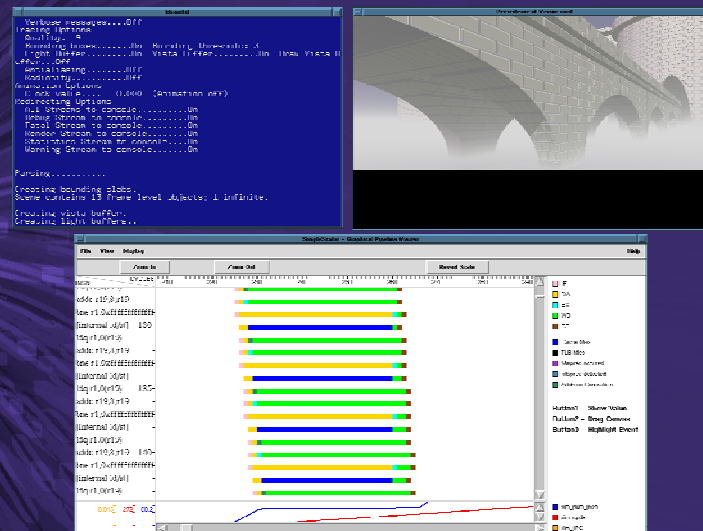
- Addresses difficulties of creating reproducible experiments
 - 1000x compression over industry-standard branch tracing
 - Re-creates of all program data values (unlike branch tracing)
- External I/O events traced
 - System calls, external interrupts
 - All events time-stamped with processor cycle count



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■ Innovation: GPV – Graphical Pipeline Viewer

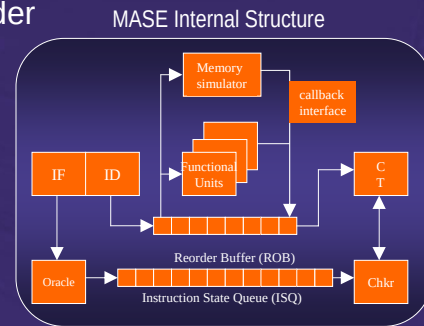


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■ Innovation: MASE Microarchitecture Simulator

- Performance simulation model more detailed than sim-outorder
- Many simulation advances
 - x86 target and uArch support
 - Improves validation support
 - Oracle for “perfect” studies
 - Speculative state management facilities simplify speculation
 - Callback interface permits sophisticated memory simulation

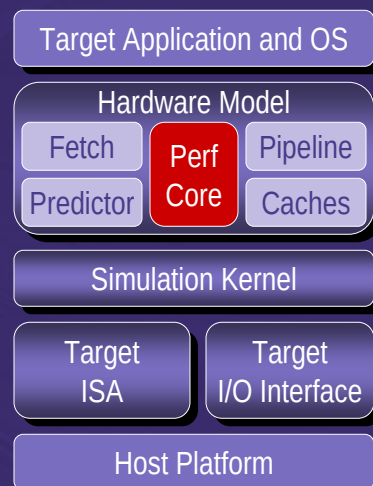


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■ Advice: Build a Platform

- Build a tool for building tools
 - “Separate mechanism and policy”
- Assume users will utilize components a-la-carte
- Strive to make the researchers code only **glue** that binds components



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Warning: Flexibility vs. Barrier-to-Entry

- A single file describes all aspects of the architecture
 - Used to generate decoders, dependency analyzers, functional components, disassemblers, appendices, etc.
 - Machine definition + 10 line main == functional simulator

```
DEFINST(ADDI,                                0x41, ← opcode
        "addi",                               "t,s,i", ← inst flags
        IntALU,                               F_ICOMP|F_IMM,
        GPR(RT), NA,                          GPR(RS), NA, NA ← input deps
        SET_GPR(RT, GPR(RS)+IMM))
assembly template → "addi",
FU req's → IntALU,
output deps → GPR(RT), NA,
semantics → SET_GPR(RT, GPR(RS)+IMM))
```

- Optimizing performance has similar effects

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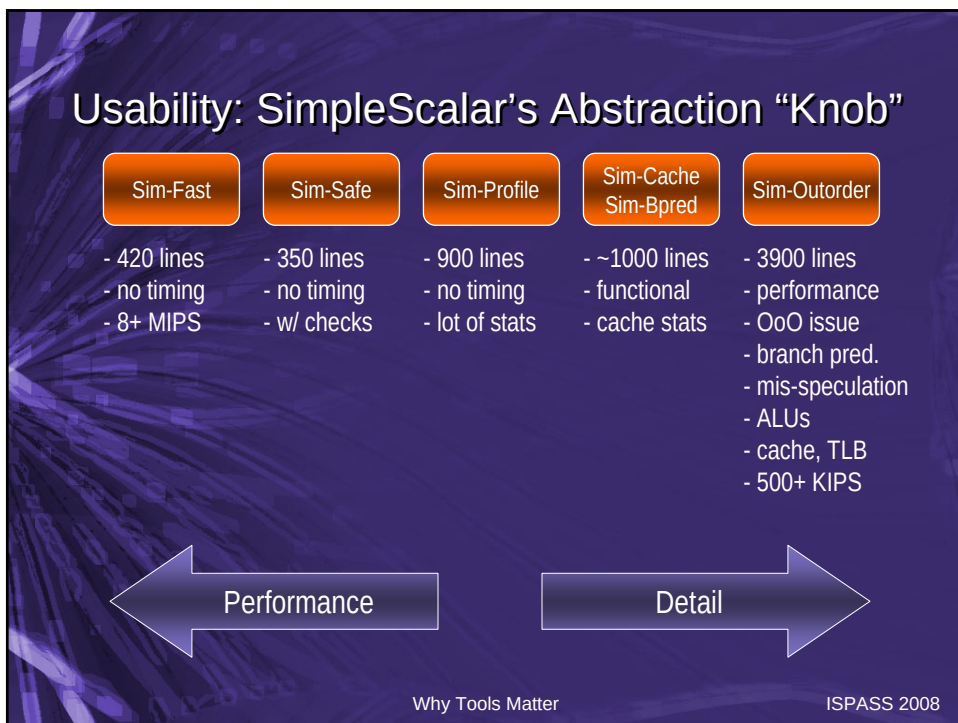
Advice: Make Usability a Priority

- Make it work out of the box
 - Include everything the user needs
 - Lots of pre-built simulators
 - Cross-compilers
 - Workload
 - Documentation and tutorials
- Open source is a must-have!
- Build portable code
- Be your first and best customer



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Advice: Work at Gaining Market Share

- Be a champion
 - Centralize decisions and integration
 - End result will be more coherent, focused
- Strive for quality, but avoid perfection
 - “Be prepared to throw one away.”
 - Build a regression test suite
 - Incorporate debugging features
- Listen to users
 - Understand how the current infrastructure limits their work
 - Track users with an easy-to-google name



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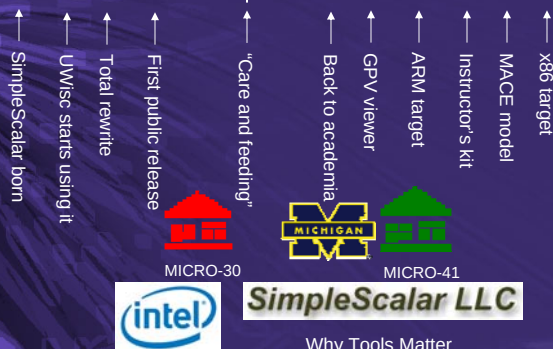
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SimpleScalar Timeline

SimpleScalar Citations (as per Google Scholar)

3 13 24 53 61 100 186 284 327 442 493 571 565 420

'93 '94 '95 '96 '97 '98 '99 '00 '01 '02 '03 '04 '05 '06 '07 '08 '09



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■ Advice: Don't Ignore Business Considerations

- Use a license that meets your needs
 - Users often confuse “open” with “free”
 - See www.OpenSource.org for licenses
- Secure university rights early
 - Before the lawyers smell the money!
- Be aware of intellectual property issues
 - More difficult to accept contributions
 - Get trademarks, web addresses, and copyrights ASAP



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■ Tomorrow's Tools

- First, look at some of today's successes:
 - SimpleScalar – uniprocessor uArch
 - Trimaran – VLIW architectures
 - Wattch – power modeling
 - HotSpot – thermal modeling
 - CACTI – cache modeling
 - GEMS, M5, RSIM – multiprocessor simulation
 - PIN – binary instrumentation
- Their recipe for success?
- Some research arenas that need more tools...
 - Reliability, Security, Application-Specific Processors, GPUs, Sensor Networking, Virtualization, Nano-technologies, Non-traditional computation models

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Summary

- The “Why’s” far outweigh the “Why Not’s”
- Six tips for undertaking tool development and release
 - Let Research Drive the Outcome
 - Be Innovative
 - Build a Platform
 - Make Usability a Priority
 - Work at Gaining Market Share
 - Don’t Ignore Business Considerations

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